An n-Valued Logic System with Graded Implication and Algebraic Contradiction Expulsion

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Abstract

This system was first formulated on 13 September 2025. We present an n-valued logic system constructed over a symmetric integer-valued truth set excluding zero, in which classical binary logic is recovered exactly as the special case $n = 2$. The system defines negation as arithmetic sign-inversion, conjunction and disjunction as order-theoretic operators, and introduces a graded implication validity function that quantifies inferential strength continuously. Contradictions are not excluded axiomatically; instead, they are ruled out by the algebraic structure of the system: the assumption of simultaneous truth and negation forces membership at zero, which is excluded from the even-n truth set by construction. A natural extension to odd n introduces a zero state representing pure uncertainty, defined axiomatically as self-negating and bidirectionally universal (both antecedent and consequent). We discuss the system's structural properties and its potential as a foundation for modelling graded human reasoning.

1 Introduction

Binary logic admits exactly two truth values—true and false—and defines the logical universe around their interaction. The widespread reach of this choice, from electronic circuits to mathematical proof, is well established. Yet when one attempts to model human decision-making by the same binary reduction, the framework becomes strained. Human judgment involves degrees of belief, partial evidence, and graded confidence that resist collapse into a single bit without losing information that is often essential.

Motivated by the question of how binary logic could be extended to an arbitrary number of states while preserving its essential character, we constructed the system described here independently from first principles. The requirement that $n = 2$ recover classical binary logic exactly served as a guiding consistency constraint throughout, and the system was designed so that this recovery is not approximate but perfect.

The resulting system is parameterised by a natural number n, defined over a symmetric integer set, with negation as sign-inversion, conjunction and disjunction as order-theoretic operations, a graded implication validity function, and algebraic—rather than axiomatic—expulsion of contradiction.

2 The Truth Set

Definition 1 (Index and Parameter)**.** *Let $n \in \mathbb{N}$ with n even. Define $k = n/2$.*

Definition 2 (Truth Set — Even n)**.** *The truth set S_n is defined as:*

$$S_n = \{-k,\ -(k-1),\ \ldots,\ -1,\ 1,\ \ldots,\ (k-1),\ k\}$$

so that $|S_n| = n$ and $0 \notin S_n$.

Definition 3 (Truth Ordering)**.** *Elements of S_n are ordered by the standard integer ordering. The element $-k$ is designated* absolute true *and $+k$ is designated* absolute false*. Intermediate values represent graded truth states.*

Definition 4 (φ Mapping)**.** *Define $\varphi : S_n \to [0,1]$ by:*

$$\varphi(x) = \frac{k - x}{2k}$$

This maps $-k \mapsto 1$ (absolute true) and $k \mapsto 0$ (absolute false). The mapping is strictly decreasing and never attains $\frac{1}{2}$, since that would require $x = 0 \notin S_n$.

Example 1. *For $n = 4$: $k = 2$, $S_4 = \{-2, -1, 1, 2\}$. Then $\varphi(-2) = 1$, $\varphi(-1) = 3/4$, $\varphi(1) = 1/4$, $\varphi(2) = 0$.*

3 Logical Operations

Definition 5 (Negation)**.** *For $A \in S_n$:*

$$\neg A = -A$$

Under φ, this satisfies $\varphi(\neg A) = 1 - \varphi(A)$.

Definition 6 (Conjunction and Disjunction)**.**

$$A \wedge B = \max(A, B), \qquad A \vee B = \min(A, B)$$

Since more negative values represent greater truth, max *selects the less true of two values, consistent with classical conjunction being limited by its weakest conjunct.*

Definition 7 (Implication)**.** *The implication $A \implies B$ is valid if and only if $A \geq B$.*

Definition 8 (Graded Implication Validity)**.** *Define $v : S_n \times S_n \to [0,1]$ by:*

$$v(p,q) = \begin{cases} 1 & \text{if } p = 0 \text{ or } q = 0 \text{ (in odd } n\text{)} \\ \dfrac{1}{1 + |p - q|} & \text{if } p \geq q \text{ and } p \neq 0, q \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

Remark 1. *$v(p,q) = 1$ if and only if $p = q$ (for $p, q \neq 0$), or if either p or q equals 0 in odd n. Validity decreases monotonically as $|p - q|$ increases for standard elements. Invalid implications return exactly zero.*

4 Key Theorems

4.1 Contrapositive Preservation

Theorem 1 (Contrapositive)**.** *For all $p, q \in S_n$: $v(p, q) > 0 \iff v(\neg q, \neg p) > 0$.*

Proof. For standard elements where $p, q \neq 0$, $v(p, q) > 0$ requires $p \geq q$. Since $\neg q = -q$ and $\neg p = -p$, multiplying by -1 reverses the inequality to give $\neg p \leq \neg q$, which is equivalently written as $\neg q \geq \neg p$, so $v(\neg q, \neg p) > 0$.

For the odd-n zero state: if $p = 0$, then $v(0, q) = 1 > 0$ for all $q \in S_n$. Since $\neg 0 = 0$ by definition, we have $v(\neg q, 0) = 1 > 0$ for all q (as zero is universally consequent: everything implies zero). Similarly, if $q = 0$, then $v(p, 0) = 1 > 0$ and $v(0, \neg p) = 1 > 0$ by the same property. Thus the equivalence holds for the zero state in both directions. The converse is symmetric. □ □

4.2 Algebraic Contradiction Expulsion

Definition 9 (Equivalence Validity Function)**.** *To quantify the relationship between two truth states $x, y \in S_n$, define $\chi : S_n \times S_n \to [0, 1]$ by:*

$$\chi(x, y) = \begin{cases} 0 & \text{if } x = -y \text{ and } x \in \{-k, k\} \\ \dfrac{1}{1 + |x - y|} & \text{otherwise} \end{cases}$$

The case $x = -y$ (given x is absolute true or false) represents a perfectly invalid equality, confirming that the simultaneous assertion of absolute true value and absolute false value is a null state in the system.

Theorem 2 (Contradiction Expulsion)**.** *For even n, there exists no element $s \in S_n$ such that the equality $s = \neg s$ possesses any degree of validity $\chi > 0$.*

Proof. Suppose $s = \neg s$ for some $s \in S_n$. By definition $\neg s = -s$, so $s = -s$, which implies $s = 0$. But $0 \notin S_n$ for even n. Furthermore, setting $x = s$ and $y = \neg s = -s$ gives $x = -y$ identically, so $\chi(s, -s) = 0$ for all $s \in S_n$. Therefore no valid contradiction exists. □ □

Remark 2. *Contradiction is not forbidden by an axiom imposed on the system—it is structurally impossible within it. The exclusion of zero is the sole load-bearing condition.*

4.3 Classical Logic Recovery

Theorem 3 (Binary Recovery)**.** *For $n = 2$, the system reduces exactly to classical binary logic.*

Proof. For $n = 2$: $k = 1$, $S_2 = \{-1, 1\}$. Designate $-1 = \top$ (True) and $1 = \bot$ (False).

Negation: $\neg\top = 1 = \bot$ and $\neg\bot = -1 = \top$. ✓

Conjunction: $\top \wedge \top = \max(-1, -1) = \top$, $\top \wedge \bot = \max(-1, 1) = \bot$, $\bot \wedge \bot = \max(1, 1) = \bot$. ✓

Disjunction: $\top \vee \top = \min(-1, -1) = \top$, $\top \vee \bot = \min(-1, 1) = \top$, $\bot \vee \bot = \min(1, 1) = \bot$. ✓

Implication: $\top \implies \top$: $-1 \geq -1$ ✓; $\bot \implies \top$: $1 \geq -1$ ✓; $\top \not\iff \bot$: $-1 \not\geq 1$ ✓; $\bot \implies \bot$: $1 \geq 1$ ✓.

All classical truth tables are recovered exactly. □ □

5 The Odd-n Extension: Pure Uncertainty

Definition 10 (Truth Set — Odd n)**.** *For odd n, define $k = (n-1)/2$ and:*

$$S_n = \{-k,\ \ldots,\ -1,\ 0,\ 1,\ \ldots,\ k\}$$

so that $|S_n| = n$ and $0 \in S_n$.

Definition 11 (Pure Uncertainty State)**.** *The element $0 \in S_n$ is the* pure uncertainty *state, defined by the following axioms:*

(i) $\neg 0 = 0$ *(by definition, not algebraic consequence)*

(ii) $v(0, q) = 1$ *and* $v(q, 0) = 1$ *for all* $q \in S_n$ *(zero is universally antecedent and consequent)*

(iii) $\varphi(0) = \frac{1}{2}$

Remark 3. *The designation $\neg 0 = 0$ as definitional rather than consequential is a necessary distinction. Algebraically $-0 = 0$ holds, but treating this as a definition makes explicit that zero occupies a categorically different ontological position: it is not a truth value on the true-false axis but a state that lies outside that axis entirely. Pure uncertainty implying everything is epistemically well-motivated—a state of complete uncertainty is compatible with any outcome and therefore blocks no inferential path.*

6 Future Directions

The discrete formulation presented here naturally suggests several extensions. A natural completion of this program would be a *continuous-state logic* in which truth values range over the real interval $[-1, 1] \setminus \{0\}$ (or its closure), negation remains sign-inversion, and the implication validity function is extended to a continuous domain. Such a system would preserve all structural properties established here—contradiction expulsion, contrapositive preservation, binary recovery as a limiting case—while removing the discretisation constraint. Future work may investigate this continuous extension.

A secondary direction of interest is the application of graded implication validity to verification problems in formal systems, particularly contexts where inferential confidence must be represented and propagated rather than collapsed to a binary judgment.

7 Conclusion

We have presented a complete n-valued logic system defined over a symmetric integer set, with negation as sign-inversion, conjunction and disjunction as order-theoretic operations, and a graded implication validity function. The system recovers classical binary logic exactly at $n = 2$, expels contradiction by algebraic necessity rather than axiom, and extends naturally to odd n through an axiomatically defined pure uncertainty state. This work provides a foundation for the continuous-state extension discussed above.